5.1.1 Of Gaussian elimination and LU factorization

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05.1.1 Gaussian elimination versus LU factorization, Part 1

fit width

Homework 5.1.1.1. Reduce the appended system

$$\begin{array}{rrr|r} 2 & -1 & 1 & 1 \\ -2 & 2 & 1 & -1 \\ 4 & -4 & 1 & 5 \end{array}$$

to upper triangular form, overwriting the zeroes that are introduced with the multipliers.

▾ Solution

$$\begin{array}{rrr|r} 2 & -1 & 1 & 1 \\ -1 & 1 & 2 & 0 \\ 2 & -2 & 3 & 3 \end{array}$$

05.1.1 Gaussian elimination versus LU factorization, Part 2

fit width

$A = \mathrm{LU}(A)$

$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$

A_{TL} is 0×0

while $n(A_{TL}) < n(A)$

$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$

$a_{21} := a_{21}/\alpha_{11}$

$A_{22} := A_{22} - a_{21} a_{12}^T$

$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{cc|c} A_{00} & a_{01} & A_{02} \\ a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array} \right)$

endwhile

Figure 5.1.1.1. Algorithm that overwrites A with its LU factorization.

Homework 5.1.1.2. The execution of the LU factorization algorithm with

$$A = \begin{pmatrix} 2 & -1 & 1 \\ -2 & 2 & 1 \\ 4 & -4 & 1 \end{pmatrix}$$

in the video overwrites A with

$$\begin{pmatrix} 2 & -1 & 1 \\ -1 & 1 & 2 \\ 2 & -2 & 3 \end{pmatrix}.$$

Multiply the L and U stored in that matrix and compare the result with the original matrix, let's call it $\widehat{A}$.

▼ Solution

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -2 & 1 \end{pmatrix} \text{ and } U = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix}.$$

$$LU = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -2 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 1 \\ -2 & 2 & 1 \\ 4 & -4 & 1 \end{pmatrix} = \widehat{A}.$$